values to *P* and *Q*; but in that case one would be true and the other would be false, and classical logic would be fine.

Despite the fact that classical logic is perfectly valid whenever it is properly applied, quantum mechanics has inspired more general approaches known as *quantum logic*, pioneered by John von Neumann and his collaborator Garrett Birkhoff. By starting with slightly different logical axioms from the standard ones, we can derive a set of rules obeyed by the probabilities implied by the Born rule in quantum mechanics. Quantum logic in this sense is both interesting and useful, but its existence does not invalidate the correctness of ordinary logic in appropriate circumstances.

○ ○ ○

Niels Bohr, in an attempt to capture what makes quantum theory so unique, proposed the concept of *complementarity*. The idea is that there can be more than two ways of looking at a quantum system, each of them equally valid, but with the property that you can't employ them simultaneously. We can describe the wave function of a particle in terms of either position or momentum, but not both at the same time. Similarly, we can think of electrons as exhibiting either particle-like or wave-like properties, just not at the same time.

Nowhere is this feature made more evident than in the famous double-slit experiment. This experiment wasn't actually performed until the 1970s, long after it was proposed. It wasn't one of those surprising experimental results that theorists had to invent a new way of thinking in order to understand, but rather a thought experiment (suggested in its original form by Einstein during his debates with Bohr, and later popularized by Richard Feynman in his lectures to Caltech undergraduates) meant to show the dramatic implications of quantum theory.

The idea of the experiment is to home in on the distinction between particles and waves. We start with a source of classical particles (maybe a pellet gun that tends to spray in somewhat unpredictable directions), shoot them through a single thin slit, then detect them at a screen on the other side of the slit. Mostly the particles will pass right through, with perhaps very slight deviations if they bump up against the sides of the slit. So what we